Constituents of atomic nucleus

All the atomic nuclei are made up of elementary particles called protons and neutrons. A proton has a positive charge of the same magnitude as that of an electron and mass of proton is $1.67 \times 10^{-27} kg$. A neutron is electrically neutral and mass of neutron is same as that of proton.

An atomic nucleus is represented by ${}^{A}_{Z}\mathrm{X}$.

Atomic number: The number of protons in a nucleus is called the atomic number. It is denoted by Z. The identity of an element depends on the atomic number in a nucleus.

Mass number: The total number of protons and neutrons in a nucleus is called mass number. It is denoted by A.

We can write $A = Z + N$

Where, Z = Number of protons
N = Number of neutrons.

Nucleon: The combined (or common) name of proton and neutron which construct nucleus is called nucleon.

Nuclide: The similar nuclei which are specified by the mass number and atomic number are known as nuclide. For example, *different isotopes of the same elements are different nuclide.*

Classification of nucleus

Isotopes: Nuclei having same number of protons but different number of neutrons are called isotopes.

Examples: ${}^{1}_{1}\mathrm{H}, {}^{2}_{1}\mathrm{H}\ and\ {}^{3}_{1}\mathrm{H}$ are isotopes of Hydrogen.

${}^{12}_{6}\mathrm{C}, {}^{13}_{6}\mathrm{C}\ and\ {}^{14}_{6}\mathrm{C}$ are isotopes of Carbon.

Isobars: Nuclei having same mass number but different atomic number/proton number are called isobars.

Examples: ${}^{11}_{4}\mathrm{Be}, {}^{11}_{5}\mathrm{B}\ and\ {}^{11}_{6}\mathrm{C}$ are isobars.

${}^{16}_{8}\mathrm{O}\ and\ {}^{16}_{7}\mathrm{N}$ are isobars.

Isotones: Nuclei having same number of neutrons but different atomic number/proton number are called isobars.

Examples: ${}^{3}_{1}\mathrm{H}, {}^{4}_{2}\mathrm{He}\ and\ {}^{5}_{3}\mathrm{Li}$ are isotones.

${}^{14}_{6}\mathrm{C}, {}^{15}_{7}\mathrm{N}\ and\ {}^{16}_{8}0$ are isotones.

Mirror nucleus: Nuclei having the same mass number but the proton and neutron number are interchanged are called mirror nucleus.

Examples: ${}^{7}_{4}\mathrm{Be}\ (\mathrm{Z} = 4, \mathrm{N} = 3)\ and\ {}^{7}_{3}\mathrm{Li}\ (\mathrm{Z} = 3, \mathrm{N} = 4)$ are mirror nucleus.

${}^{15}_{8}\mathrm{O}\ (\mathrm{Z} = 8, \mathrm{N} = 7)\ and\ {}^{15}_{7}\mathrm{N}\ (\mathrm{Z} = 7, \mathrm{N} = 8)$ are mirror nucleus.

Measurement of nuclear mass

In various phenomenona of nucleus it is seen that mass is converted into energy and vice-versa. So, it is convenient to express mass and energy in same unit.

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If m kg is the mass of the particle then the equivalent energy of the mass is mc^2 Joule.

We know

$$E = mC^2$$

$$\therefore \qquad E = m \times (2.9979 \times 10^8)^2 \qquad Joule$$

One electron volt unit is defined as the amount of work an electron does to cross a potential difference of 1 volt.

Therefore

$$1\ eV = 1.6022 \times 10^{-19} Joule$$

$$1\ MeV = 10^6 eV = 1.6022 \times 10^{-13} Joule$$

Therefore the equivalent energy of m kg is

$$E = m \times (2.9979 \times 10^8)^2 \qquad Joule$$

$$E = m \times 8.9874 \times 10^{16} \qquad Joule$$

$$E = m \times \frac{8.9874 \times 10^{16}}{1.6022 \times 10^{-19}} \qquad eV$$

$$E = m \times 5.6095 \times 10^{35} \qquad eV$$

$$\therefore \qquad E = m \times 5.6095 \times 10^{29} \qquad MeV$$

We know, One atomic mass (1 amu) is taken as one-twelfth (1/12) of the mass of carbon atom $_6C^{12}$ and

$$1\ a.m.u = \frac{1}{12}(Mass\ of\ Carbon - 12\ atom)$$

We know in 1 gm-atom the number of atoms is 6.022×10^{23}. This is known as Avogadro's number. So, in 12 gm Carbon there are 6.022×10^{23} atoms of Carbon.

So, the mass of a Carbon-12 atom

$$= \frac{12}{6.022 \times 10^{23}} = 1.99269 \times 10^{-23} g$$

Thus, atomic mass unit

$$1\ a.m.u = \frac{1}{12} \times 1.99269 \times 10^{-23} g$$

$$1\ a.m.u = 1.66057 \times 10^{-27} kg$$

$$1\ a.m.u = (1.66057 \times 10^{-27} \times 5.6095 \times 10^{29}) \quad MeV$$

$$\therefore \qquad 1\ a.m.u = 931.5 \quad MeV$$

Radius of the nucleus: The size of the nucleus depends upon the number of protons and the number of neutrons inside it. Thus the volume of a nucleus is directly proportional to the number of nucleons contained in it, which is its mass number A.

If R is the nuclear radius, then $\qquad V \propto A$

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$$\frac{4}{3}\pi R^3 \propto A$$

$$R^3 \propto A$$

$$R \propto A^{1/3}$$

$$\therefore \quad R = R_0 A^{1/3}$$

Where R_0 is a constant and $R_0 \cong 1.2 \times 10^{-15} m$.

Nuclei are so small that the unit of length appropriate in describing them is the femtometer (*fm*) equal to $10^{-15}\ m$. The femtometer often called *Fermi*.

Thus the radius of the ${}^{27}_{13}\mathrm{Al}$ nucleus is

$$R = 1.2 \times 10^{-15} \times (27)^{1/3}$$

$$\therefore \quad R = 3.6\, fm$$

Mass defect: Neutrons and protons together create a nucleus. So, the mass of a nucleus should be equal to the mass of the protons and neutrons added together. But actually the mass of a nucleus is little less than the mass of the protons and neutrons added together. The difference in mass is known as mass defect. It is denoted by *ΔM*.

Mathematically, $\Delta M = ZM_p + (A-Z)M_n - M$

Where, Z = Number of protons
$(A\text{-}Z)$ = Number of neutrons.
M_p = Mass of protons
M_n = Mass of neutrons
M = Mass of nucleus.

Binding energy: The energy equivalent to mass defect is called the binding energy of the nucleus. If we want to break the nucleus into separate protons and neutrons, again we have to introduce the same amount of energy that is lost during creation. *The minimum amount of energy needed to break a nucleus into its constituents neutrons and protons, is called the binding energy of the nucleus.*

The relation between the binding energy and mass defect is given by

$$B.E. = \Delta MC^2$$

$$B.E. = \{ZM_p + (A-Z)M_n - M\}C^2$$

If all masses are in atomic mass unit (a.m.u.) then

$$B.E. = \{ZM_p + (A-Z)M_n - M\} \times 931.5\, MeV$$

The binding energy is always a positive quantity.

Nuclear reaction

When a nucleus gets in close contact with another nucleus, the incident particle and the target nucleus form a composite system which is an excited state and after a short while a reaction is produced in which the incident particle itself or some other particle or gamma ray is emitted with excess energy and a resulting nucleus is obtained. This phenomenon is called a nuclear reaction.

A nuclear reaction can be written as

$$a + X \rightarrow Y + b$$

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$$Or \qquad\qquad X(a,b)Y$$

Where, *a* is the incident particle
X is the target nucleus
Y is the residual product nucleus
b is the resulting or outgoing particle.

a and *b* are incident and outgoing particle respectively and these may be any one of the following: Proton$\left({}_1^1\mathrm{H}\right)$, Neutron$({}_0^1\mathrm{n})$, Deuteron$\left({}_1^2\mathrm{H}\right)$, Alpha particle$\left({}_2^4\mathrm{He}\right)$, Gamma ray$(\gamma)$ or a few nucleons.

Examples:

Nuclear reaction	***Short form of Nuclear Reaction***
${}_7^{14}\mathrm{N} + {}_2^4\mathrm{He} \rightarrow {}_8^{17}\mathrm{O} + {}_1^1\mathrm{H}$	${}_7^{14}\mathrm{N}(\alpha,\mathrm{p}){}_8^{17}\mathrm{O}$
${}_3^6\mathrm{Li} + {}_1^2\mathrm{H} \rightarrow {}_4^7\mathrm{Be} + {}_0^1\mathrm{n}$	${}_3^6\mathrm{Li}(\mathrm{d},\mathrm{n}){}_4^7\mathrm{Be}$
${}_{17}^{35}\mathrm{Cl} + {}_1^1\mathrm{H} \rightarrow {}_{16}^{32}\mathrm{S} + {}_2^4\mathrm{He}$	${}_{17}^{35}\mathrm{Cl}(\mathrm{p},\mathrm{n}){}_{16}^{32}\mathrm{S}$
${}_5^{10}\mathrm{B} + {}_2^4\mathrm{He} \rightarrow {}_7^{13}\mathrm{N} + {}_0^1\mathrm{n}$	${}_5^{10}\mathrm{B}(\alpha,\mathrm{n}){}_7^{13}\mathrm{N}$
${}_7^{14}\mathrm{N} + {}_0^1\mathrm{n} \rightarrow {}_6^{13}\mathrm{C} + {}_1^2\mathrm{H}$	${}_7^{14}\mathrm{N}(\mathrm{n},\mathrm{d}){}_6^{13}\mathrm{C}$
${}_{12}^{26}\mathrm{Mg} + {}_1^1\mathrm{H} \rightarrow ({}_{13}^{27}\mathrm{Al})^* \rightarrow {}_{13}^{27}\mathrm{Al} + \gamma$	${}_{12}^{26}\mathrm{Mg}(\mathrm{p},\gamma){}_{13}^{27}\mathrm{Al}$

Nuclear fission and fusion reaction

Nuclear fission: If neutrons or gamma rays of definite speed strike a heavy nucleus (A>230) and breaks the nucleus into two main parts, then a vast nuclear energy is released. This kind of breakup of nucleus is called nuclear fission.

Let us consider the capture of neutron by a heavy nucleus ${}_{92}^{235}\mathrm{U}$. The reaction can be represented as

$${}_{92}^{235}\mathrm{U} + {}_0^1\mathrm{n} \rightarrow ({}_{92}^{236}\mathrm{U})^* \rightarrow {}_{56}^{141}\mathrm{Ba} + {}_{36}^{92}\mathrm{Kr} + 3{}_0^1\mathrm{n} + \mathrm{Energy}\ (\cong 200\ \mathrm{MeV})$$

Nuclear fusion: More than one light nuclei are fused (or combined) together to form a single heavy nucleus and produce enormous nuclear energy. This process is called nuclear fusion.

The reaction can be represented as

$${}_1^2\mathrm{H} + {}_1^3\mathrm{H} \rightarrow {}_2^4\mathrm{He} + {}_0^1\mathrm{n} + \mathrm{Energy}\ (\cong 17.6\ \mathrm{MeV})$$

Differences between nuclear fission and fusion reaction: The differences between nuclear fission and fusion reaction are given below:

Nuclear fission	***Nuclear fusion***
In this process heavy nucleus is splitted into lighter nucleus.	In this process lighter nuclei are combined to produce heavy nucleus.
Energy released is large.	Energy released is small.
${}_{92}^{235}\mathrm{U}$, ${}_{94}^{239}\mathrm{Pu}$ nuclei are used.	${}_1^2\mathrm{H}$, ${}_1^3\mathrm{H}$, ${}_2^4\mathrm{He}$ nuclei are used.
In this process radioisotopes are produced.	In this process radioisotopes are not produced.
The link of the fission process is with neutrons.	The link of the fusion process is with protons.

1. i) ${}_0^1n + {}_{92}^{235}U \rightarrow \left[{}_{92}^{236}U\right] \rightarrow {}_{56}^{141}Ba + {}_{36}^{92}Kr + 3{}_0^1n + E_1$

   ii) ${}_1^2H + {}_1^2H \rightarrow {}_2^4He + E_2$

   Where, $M({}_{92}^{235}U) = 235.0439\,amu$

   $M({}_{56}^{141}Ba) = 140.9139\,amu$

   $M({}_{36}^{92}Kr) = 91.8973\,amu$

   $M({}_1^2H) = 2.01478\,amu$

   $M({}_2^4He) = 4.00388\,amu$

   $M({}_0^1n) = 1.0087\,amu$

   Find E_1 & E_2 of Equations (I & II).

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